



Cultivation of *Spirulina platensis* for nutrient removal from piggery wastewater

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Abstract

The discharge of livestock wastewater without appropriate treatment causes severe harm to the environment and human health. In the pursuit of finding solutions to this problem, the cultivation of microalgae as feedstock for biodiesel and animal feed additive using livestock wastewater coupled with the removal of nutrients from wastewater has become a hot research topic. In this study, the cultivation of *Spirulina platensis* using piggery wastewater for the production of biomass and the removal of nutrients were studied. The results of single factor experiments confirmed that Cu^{2+} seriously inhibit the growth of *Spirulina platensis*, while the influences of nitrogen, phosphorous, and zinc on the growth of *Spirulina platensis* can all be described as “low promotes high inhibits.” *Spirulina platensis* grew well in the 4-fold dilution of piggery wastewater supplemented with moderate sodium bicarbonate, which indicated that it is the limiting nutrients for *Spirulina platensis* growth in piggery wastewater. The biomass concentration of *Spirulina platensis* reached 0.56 g/L after 8 days of culture at the optimal conditions proposed by the response surface method, which were as follows: 4-fold dilution of piggery wastewater, 7 g/L sodium bicarbonate, pH of 10.5, initial OD_{560} of 0.63, light intensity of 3030 lx, and light time/dark time of 16 h/8 h. *Spirulina platensis* cultured in the diluted piggery wastewater contained 43.89% protein, 9.4% crude lipid, 6.41 mg/g chlorophyll a, 4.18% total sugar, 27.7 mg/kg Cu, and 246.2 mg/kg Zn. The removal efficiency for TN, TP, COD, Zn, and Cu from the wastewater by *Spirulina platensis* was 76%, 72%, 93.1%, 93.5%, and 82.5%, respectively. These results demonstrated the feasibility of piggery wastewater treatment by the cultivation of *Spirulina platensis*.

Keywords Piggery wastewater · *Spirulina platensis* · Nutrient removal · Wastewater treatment · Chemical composition

Introduction

The breeding industry is rapidly growing in response to economic growth and expanding urbanization. The live pig production in the country was over 544 million according to the China Statistical Yearbook 2019, and the piggery wastewater output was about 6.8 million tons/days in 2010 (Chen et al. 2017; China 2019). Piggery wastewater contains a high concentration of nitrogen, phosphorus, chemical

oxygen demand (COD), and a number of heavy metals, which will harm the environment and human health in the lack of prompt wastewater treatment (Greenstein et al. 2018; Li et al. 2019).

The cultivation of microalgae in livestock wastewater for the removal of nutrients has become a hot research topic due to advantages such as highly efficient nutrient removal (mainly N and P), low cost, eco-friendliness, and the improvement of atmosphere quality through the biotransformation of CO_2 to O_2 (Ganeshkumar et al. 2018; Gupta et al. 2016; Mousavi et al. 2018; Mulbry et al. 2008; Nam et al. 2016). Recently, many studies have focused on the treatment of livestock wastewater by the cultivation of microalgae to produce biomass as feedstock for biodiesel, and *Chlorella vulgaris* have been commonly used for its high tolerance for nitrogen and high fatty acid content (Gupta et al. 2016; Nam et al. 2016; Qu et al. 2021; Zheng et al. 2018). It is noteworthy that the commercial application of this strategy

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is hampered by the high cost of harvesting the biomass, which commonly exceeds 30% the total cost of microalgae production (Alam et al. 2015; Smith and Davis 2012; Zhou et al. 2014).

One of the preconditions of the commercial application of cultivation of microalgae in livestock wastewater for the removal of nutrients is that the harvesting of microalgae must be economical and simple. *Spirulina platensis* (*S. platensis*), a photosynthetic cyanobacterium, has been successfully produced commercially as the feedstock for healthy foods and animal feed, which mainly for its easy process of cultivation, relatively high growth rate, and can be harvested by simple filtration (Papadaki et al. 2017; Rodrigues et al. 2010).

Some researchers have focused on the removal of nutrients from wastewater by the cultivation of *S. platensis*. Chang et al. (2013) reported that, in an autotrophic culture of *S. platensis* with a dilution of human urine, the removal efficiency for NH_4^+ -N, total phosphorus, and urea was 97%, 96.5%, and 85–98%, respectively. They also found that the mixotrophic cultivation of *S. platensis* in human urine supplemented with adequate amounts of organic carbon sources is a promising method to produce microalgae biomass as a biofuel feedstock. Markou et al. (2012) found that cultivating *S. platensis* in olive-oil mill wastewater removed 73.18% of COD and 91.19% of carbohydrates. Feng and Wu (2006) found that *S. platensis* can be cultivated in the diluted urine solution for the production of biomass coupled with O_2 production. A number of studies have focused on the cultivation of *S. platensis* in urine solution/olive-oil mill wastewater to remove N and P and produced biomass as the feedstock of biofuel. However, the treatment of piggery wastewater by the cultivation of *S. platensis* and the characterization of the chemical composition of biomass has received little attention. Meanwhile, the feasibility for the cultivation of *S. platensis* in piggery wastewater for the production of animal feed has been rarely reported.

In the present study, we evaluated the influence of different components in the wastewater as well as the effect of adding sodium bicarbonate to diluted piggery wastewater on the growth of *S. platensis*. The cultivation parameters of *S. platensis* culture in the diluted piggery wastewater were optimized using the Response Surface Method (RSM), and the removal of nutrients, and the chemical composition of the biomass were analyzed.

Materials and methods

***S. platensis* cultivation** The *S. platensis* culture used in the present study was obtained from Jiangxi Alga Biology Technology Co., Ltd., Jiangxi Province, China. *S. platensis* was cultivated in improved Zarrouk medium (pH was

adjusted to 10.0 with 1N NaOH) in a light culture chamber at $30.0 \pm 0.5^\circ\text{C}$ and 12 h/d light until the OD_{560} exceeded 1.2. The inoculum was filtrated by using vacuum suction filtration and the biomass was washed three times with deionized water to remove the nutrient salts adsorbed on the cell surface and then suspended in moderate 1% NaCl solution.

Piggery wastewater

Piggery wastewater was collected from the supernatant of the wash water of pig manure and urine after sedimentation from the swine manure treatment plant of a pig farming enterprise of Zhumadian, Henan, China. The sample was filtrated by vacuum suction filtration, then was treated with 1% NaOCl, and the filtrate was kept at 4°C .

Cultivation of *S. platensis* and subsequent removal of nutrients from piggery wastewater The influence of ingredients (nitrogen, phosphorus, Cu^{2+} , and Zn^{2+}) on the growth of *S. platensis* was evaluated by indirectly monitoring the growth of *S. platensis* in improved Zarrouk medium (lacking the studied element) that was supplemented with different concentrations of ammonia sulfate, sodium phosphate, Cu^{2+} , and Zn^{2+} . In the present study, the concentrations of nitrogen, phosphorus, Cu^{2+} , and Zn^{2+} all referred to the element concentration. Control experiments were set with no addition of the studied ingredients into the medium or wastewater. The experiments were conducted in a 500 mL flask containing 250 mL improved Zarrouk medium. The initial OD_{560} was adjusted to about 0.4 by adding concentrated *S. platensis* solution into the medium, and the initial pH was adjusted to about 10.0 with 1N NaOH. The culture was incubated for the set time in a light culture chamber at $30.0 \pm 0.5^\circ\text{C}$, with a light intensity of 3030 lx and light time of 12 h.

The optimal dilution ratio was 4 (wastewater was added into 3-fold volume of city water), which was determined mainly based on the ingredient concentration in the piggery wastewater and the pre-cultivation of *S. platensis* in different dilute ratios of wastewater. It is well known that N and P are the dominant growth-limiting nutrients for microalgae in aquatic ecosystems (Smith et al. 2010). The N and P contained in the diluted piggery wastewater can meet the growth need of *S. platensis*. Meanwhile, one of our principal aims was to achieve the removal of nutrients (nitrogen and phosphorus) and heavy metal ions (Cu^{2+} and Zn^{2+}) from the wastewater. Therefore, the growth of *S. platensis* in the 4-fold dilution of piggery wastewater supplemented with different concentration sodium bicarbonate was monitored to evaluate the effect of this nutrient. The cultivation conditions were the same as detailed above.

Based on the above findings, 4-fold dilution of piggery wastewater was used to culture the *S. platensi*. The sodium bicarbonate concentration, light intensity, light time, and pH were selected as the culturing variables. A Box–Behnken design (BBD) in the Response Surface Method (RSM) using Design-Expert 8.06 (Stat Ease Inc., UK) was used to investigate the effects of the four independent variables on the growth of *S. platensi*. The total number of runs required in the Box–Behnken design was defined as $N = 2k(k-1) + Co$, where k denotes the number of variables studied and Co the number of central points (Bezerra et al. 2008). In the present study, 29 experimental runs were performed with 5 replicates at the center of the design to estimate the pure error sum of squares, and the OD_{560} as response. The experimental design matrix in coded and uncoded units is given in Table 1. RSM experiments were conducted in the 500 mL flask containing 250 mL 4-fold dilution of piggery wastewater supplemented with the set concentration of sodium

bicarbonate. The initial OD_{560} was adjusted to about 0.4 by adding concentrated *S. platensi* solution into the wastewater. The pH, light intensity, and light time were adjusted to the set values as shown in Table 1, and the cultivation time was 8 days.

The goodness of the fit of the model was evaluated by the determination coefficient (R^2), where R^2 closer to 1 means the good agreement between the experimental and predicted values. The significance of the single factor and the interactions of the two-factors on the influence of OD_{560} were characterized by the analysis of variance (ANOVA). Statistical significance was evaluated by checking whether the p -value < 0.05 , $p < 0.05$ was considered to have a significant effect on the response (Ghafari et al. 2009). The RSM experimental data used were the average of triplicates within $\pm 10\%$ standard error. Three-dimensional (3D) plots and their respective contour plots were obtained for OD_{560} based on the effects of the significant factors. The quadratic

Table 1 BBD matrix and the results of OD_{560} optimization

Run	Sodium bicarbonate concentration (g/L) A	Light intensity (Lux) B	Light time (h) C	pH D	OD_{560}	
					Measured	Predicted
1	5.50	2280	16.00	10.50	1.62	1.58
2	5.50	2280	8.00	12.00	1.25	1.19
3	5.50	2280	16.00	10.50	1.60	1.58
4	4.00	2280	16.00	9.00	1.47	1.53
5	5.50	1530	24.00	10.50	1.45	1.47
6	4.00	2280	8.00	10.50	1.27	1.31
7	5.50	2280	16.00	10.50	1.59	1.58
8	5.50	3030	24.00	10.50	1.95	2.05
9	7.00	3030	16.00	10.50	1.70	1.68
10	5.50	3030	16.00	12.00	1.68	1.66
11	7.00	2280	24.00	10.50	1.86	1.79
12	5.50	1530	16.00	12.00	1.17	1.19
13	4.00	1530	16.00	10.50	1.22	1.23
14	4.00	2280	16.00	12.00	1.49	1.48
15	5.50	2280	16.00	10.50	1.59	1.58
16	4.00	2280	24.00	10.50	1.92	1.87
17	5.50	3030	8.00	10.50	1.40	1.43
18	5.50	1530	16.00	9.00	1.19	1.18
19	5.50	2280	8.00	9.00	1.32	1.29
20	5.50	2280	16.00	10.50	1.48	1.58
21	4.00	3030	16.00	10.50	1.78	1.80
22	7.00	2280	16.00	9.00	1.41	1.46
23	7.00	2280	8.00	10.50	1.31	1.33
24	7.00	2280	16.00	12.00	1.51	1.50
25	5.50	2280	24.00	9.00	1.77	1.73
26	5.50	2280	24.00	12.00	1.78	1.74
27	7.00	1530	16.00	10.50	1.23	1.25
28	5.50	3030	16.00	9.00	1.71	1.66
29	5.50	1530	8.00	10.50	1.12	1.07

equation model was obtained by Design-Expert 8.06 which was employed to estimate the response of the dependent variable (OD_{560}) according to Eq. (1).

$$y = b_0 + \sum_{i=1}^k b_i x_i + \sum_{i=1}^k b_{ii} x_i^2 + \sum_{i=1}^{k-1} \sum_{j=i+1}^k b_{ij} x_i x_j \quad (1)$$

where y represents the predicted response (OD_{560}), x_i and x_j represent the studied varies, b_0 indicates the constant coefficient, b_i denotes the coefficient that determines the influence of variable i in the response, b_{ij} denotes the coefficient that determines the effect of interaction between variables i and j , b_{ii} is the parameter that determines the shape of the curve (quadratic effect), and k is the number of variables studied (Muralidhar et al. 2001; Zhang et al. 2012).

Analysis methods

Growth analysis

The biomass concentration of *S. platensi* was indirectly measured by spectrophotometry at 560 nm according to the regression Eq. (2) of dry weight against OD_{560} (Markou et al. 2012).

$$\text{Dry biomass (g/L)} = 0.3626OD_{560} - 0.107, R^2 = 0.994 \quad (2)$$

The pH values of culture were measured with a pH meter (PHS-3C, INESA Scientific Instrument Co., Ltd., China).

Sample preparation

After the *S. platensi* had been cultured in the dilution piggy wastewater for the set time, the biomass was harvested by vacuum suction filtration, then washed three times with deionized water to remove the inorganic salts adsorbed on the biomass surface, and lyophilized. The residual nutrients and heavy metals of the filtrate were analyzed to determine the efficiency of their removal.

Removal efficiency analysis

Total nitrogen (TN), total phosphorus (TP), and chemical oxygen demand (COD) were detected by the alkaline potassium persulfate digestion spectrophotometric method, ammonium molybdate spectrophotometric method, and dichromate titration method, respectively. The zinc and copper ion concentration were detected by inductively coupled plasma atomic emission spectrometry (ICP-MS 7800, Agilent, America). The removal efficiency (Re , %) of the nutrient and heavy metal from the wastewater was calculated according to Eq. (3).

$$Re_i = \frac{S_{i0} - S_{if}}{S_{i0}} \times 100\% \quad (3)$$

where Re_i denotes the removal efficiency (%) of the nutrients or heavy metal ions, indicated by i (TN, TP, COD, zinc and copper ions), and $i0$ and if represent the initial and final concentration of i , respectively.

***S. platensi* biomass chemical composition** The chlorophyll a content of *S. platensi* was measured by using acetone extraction-spectrophotometry according to Doucha and Lívanský (2012) and Lichtenthaler (1987). In brief, 10 mg *S. platensi* lyophilized powder was put into a glass mortar containing 10 mL 80% acetone and some quartz sand, then the biomass was extracted by grinding until the cell debris was colorless. Next, the extract was filtrated through filter paper and cleaned by 80% acetone three times. Finally, the filtrate was supplemented to a constant volume of 25 mL with 80% acetone. The absorbance of the extraction liquid was measured at 645 nm and 663 nm. The blank solution was 80% acetone. The concentration of chlorophyll a in the extraction liquid was calculated according to Eq. (4).

$$\text{Chl} - a \text{ (mg/L)} = 12.25A_{663} - 2.79A_{645} \quad (4)$$

The C, N, H, and S content of *S. platensi* biomass cultured with common medium and piggy wastewater were measured with a Vario EL cube elemental analyzer (Elementar, German). The protein content was determined from the nitrogen content measured by the elemental analyzer by using the nitrogen-to-protein conversion factor of 6.27 (Safi et al. 2013). The crude lipids of *S. platensi* were extracted and analyzed using the chloroform-methanol method. The sugar composition and content were detected by a Shimadzu LC-20AD high performance liquid chromatographic instrument equipped with C18 4.6 × 200 mm 5 μm (Shimadzu, Japan) Xtimate.

Statistical analysis

All experiments in this study were conducted in independent triplicates, and the average value was used. The relationship between the response value OD_{560} and sodium bicarbonate concentration, pH, light intensity, and light time was fitted with a quadratic equation model, and the accuracy of the model was investigated by ANOVA using Design-Expert 8.06.

Results and discussion

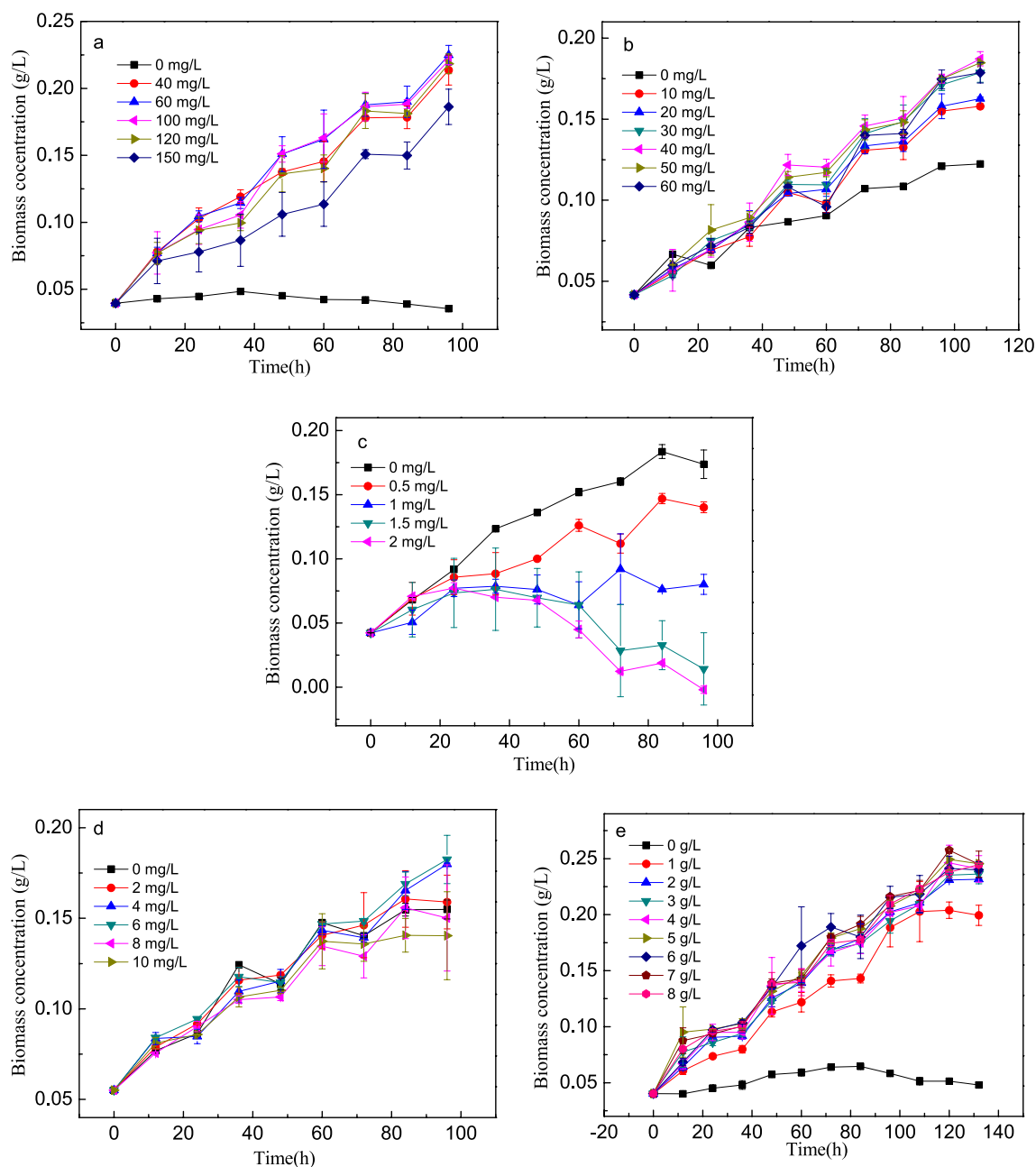
Evaluation of the influence of wastewater on the growth of *S. platensi* The concentrations of TN, TP, COD, Cu^{2+} , and Zn^{2+} in the piggy wastewater filtrate was analyzed and the results are shown in Table 2.

Table 2 Main composition of the piggery wastewater

Zn ²⁺ (mg/L)	Cu ²⁺ (mg/L)	Total nitrogen (mg/L)	Total phosphate (mg/L)	COD (mg/L)
3.01	0.39	269	97.1	3.05 × 10 ³

As shown in Fig. 1a, the growth of *S. platensi* was poor in the lack of nitrogen source in the improved Zarrouk medium, as N is the limiting nutrient for microalgae in aquatic ecosystems (Smith et al. 2010). Meanwhile, the growth of *S. platensi* was

effectively promoted by the presence of nitrogen. The biomass concentration quickly increased with the increase in nitrogen content of up to 60 mg/L, and then showed a decreasing trend when the nitrogen concentration exceeded 60 mg/L nitrogen level. The maximum biomass concentration of 0.225 g/L was achieved at 60 mg/L. The reasons for the above results are that nitrogen is mainly integrated in proteins in microalgae, and low N supply will be inhibiting the synthesis of proteins, while higher concentration also inhibits the growth of microalgae for the improper N/P ratio (Beuckels et al. 2015;

**Fig. 1** Effect of nitrogen (a), phosphorus (b), cupric ion (c), zinc ion (d), and sodium bicarbonate (e) on the growth of *S. platensi*

Chang et al. 2013). Therefore, the nitrogen concentration should not exceed 60 mg/L.

As depicted in Fig. 1b, the studied sodium phosphate concentration all significantly promoted the growth of *S. platensi*. The biomass concentration kept increasing with the increasing of phosphorus concentration, and a maximum biomass concentration of 0.187 g/L was recorded at 40 mg/L P. However, the biomass concentration recorded a decreasing trend when the phosphorus concentration was higher than 40 mg/L. This is because P is mainly integrated in the ribosomal RNA and coenzymes; thus, lower or higher supply of P will limit the synthesis of ribosomal RNA or result in the improper ratio of N to P (Beuckels et al. 2015). Therefore, the phosphorus concentration in the medium should not exceed 40 mg/L.

It can be seen from Fig. 1c, the studied concentrations of Cu^{2+} all inhibited the growth of *S. platensi*, and the inhibition degree increasing with the increasing in Cu^{2+} concentration. A biomass concentration lower than the initial value 0.042 g/L was observed when the Cu^{2+} concentration higher than 1 mg/L, along with obvious yellowing (the picture is not shown in here). This indicated that the growth of *S. platensi* ceased at a high concentration of Cu^{2+} . On the other hand, the growth of *S. platensi* was slightly inhibited at 0.5 mg/L Cu^{2+} , and the biomass was still green. These results are consistent with Li et al. (2018); they also found that the growth of both microalgae *Isochrysis galbana* and *Coelastrella* was inhibited by all studied Cu^{2+} concentrations, and stopped when it was higher than 1.0 mg/L. Therefore, the Cu^{2+} concentration of the wastewater medium should be lower than 0.5 mg/L.

Figure 1d shows that the biomass concentration elevated with the increase in Zn^{2+} concentration when it was no higher than 6 mg/L, and reached 0.182 mg/L at 6 mg/L Zn^{2+} . The result is consistent with Huang et al. (2009), they also found the growth of *Chlorella vulgaris* was prompted by the addition of Zn^{2+} , while higher concentration Zn^{2+} inhibited the growth in comparison with optimal concentration. The biomass concentration exhibited a decreasing trend when the Zn^{2+} concentration was higher than 6 mg/L. The results indicated the influence of zinc on the growth of *S. platensi* exhibited “low promote high inhibition.” Zinc is vital for the growth of microalgae through participating in the fixation of CO_2 , DNA transcription, and phosphorus acquisition (Krystian et al. 2015). Therefore, adequate amount of zinc is need to maintain the growth requirements of *S. platensi*. These results are consistent with Mata et al. (2013), who also found no obvious inhibition phenomenon in the growth of *Dunaliella tertiolecta* even under the present of $20\times\text{Zn}$ in the medium. Therefore, 6 mg/L zinc is needed to meet the growth requirements of *S. platensi*.

Figure 1e shows that *S. platensi* grew very slowly in the control experiment, and the biomass concentration slightly

increased from 0.04 to 0.06 g/L after 8 days of cultivation. In comparison with the control experiment, the maximum biomass concentration generally increased with the increase in sodium bicarbonate concentration, and a maximum value of 0.258 g/L was recorded at 7 g/L sodium bicarbonate. The promotive effect of sodium bicarbonate on the growth of *S. platensi* cultured in the dilution wastewater can be divided into a quick stage (0~4 g/L) and a slow stage (4~7 g/L). The maximum biomass concentration increased from 0.06 to 0.246 g/L in the quick stage, while it slightly increased from 0.246 to 0.258 g/L in the slow stage. The biomass concentration exhibited slight oscillation and an increasing trend when the sodium bicarbonate concentration was higher than 4 g/L. The findings showed that *S. platensi* can grow well in the diluted piggery wastewater when it was supplemented with no less than 4 g/L sodium bicarbonate. This is mainly because the necessary elements (such as nitrogen, phosphorus, and zinc ions) for the growth of *S. platensi* were provided by the diluted piggery wastewater (Zhou et al. 2014). Therefore, sodium bicarbonate is a limiting nutrient for the cultivation of *S. platensi* in diluted piggery wastewater.

As for the effects of nutrients in the piggery wastewater on the growth of *S. platensi*, nitrogen, phosphorus, and zinc all had a promotive effect, and they exhibited “low promotes high inhibits.” Cupric ions strongly inhibited the growth of *S. platensi*. Sodium bicarbonate was a limiting nutrient for the cultivation of *S. platensi* in diluted piggery wastewater.

Response surface optimization

It can be seen from Table 1 that the measured data were very close to the predicted value, which indicated that the proposed quadratic model was statistically suitable for the prediction of the growth of *S. platensi* in diluted piggery wastewater, and the RSM model in terms of the coded factors could be determined to calculate OD_{560} of *S. platensi* and it can be described by Eq. (5).

$$\begin{aligned}\text{OD}_{560} = & 1.58 - 0.011A + 0.24B + 0.26C + 0.000583D \\ & - 0.022AB - 0.025AC + 0.021AD + 0.055BC - 0.0025BD \\ & + 0.02CD - 0.015A^2 - 0.0861B^2 + 0.014C^2 - 0.067D^2\end{aligned}\quad (5)$$

where the descriptions of A, B, and C are the same as in Table 1.

The results of ANOVA for the proposed model are listed in Table 3. The p -value of the model < 0.0001 indicated that the model was robust and suitable for the prediction of *S. platensi* growth (Zhang et al. 2012). Lack-of-fit (LOF), an indicator for the influence of noise on the prediction by the model, was considered to be significant and suit for the prediction of experiment data under the conditions

Table 3 ANOVA of the fitted quadratic polynomial model of OD₅₆₀

Source	Sum of squares	df	Mean square	F value	p-value	
Model	1.55	14	0.11	32.42	<0.0001	Significant
A-Sodium bicarbonate concentration	0.00147	1	0.00147	0.43	0.5218	
B-Light intensity	0.67	1	0.67	196.81	<0.0001	
C-Light time	0.78	1	0.78	228.48	<0.0001	
D-pH	0.00000408	1	0.00000408	0.001196	0.9729	
AB	0.002025	1	0.002025	0.59	0.4541	
AC	0.0025	1	0.0025	0.73	0.4066	
AD	0.001722	1	0.001722	0.50	0.4893	
BC	0.012	1	0.012	3.54	0.0807	
BD	0.000025	1	0.000025	0.00732	0.9330	
CD	0.0016	1	0.0016	0.47	0.5049	
A ²	0.001427	1	0.001427	0.42	0.5284	
B ²	0.048	1	0.048	14.20	0.0021	
C ²	0.001189	1	0.001189	0.35	0.5645	
D ²	0.029	1	0.029	8.61	0.0109	
Residual	0.048	14	0.003415			
Lack-of-fit	0.036	10	0.003569	1.18	0.4743	not significant
Pure error	0.012	4	0.00303			
Cor total	1.60	28				

Std. dev. 0.06 R-squared 0.9678 Adj R-squared 0.9355

Pred R-squared 0.8547 Adeq precision 22.687

that LOF > 0.05 (Ghafari et al. 2009). In the present study, LOF 0.4743 > 0.05 indicated that the prediction ability of the developed model could not be disturbed by the noise. Adequate precision is the ratio value of signal to noise, and which higher than 4 is generally considered to be desirable (Ghafari et al. 2009; Mohajeri et al. 2010). The adequate precision value obtained by the model of OD₅₆₀ was 23.396, which confirmed the consistency of the modeled results and measured variables. The closer the correlation coefficient (R^2) to 1, the stronger and better the model prediction response (Mohajeri et al. 2010). Herein, the values of R^2 (0.9678), adjusted R^2 (0.9355), and predicted R^2 (0.8595) all confirmed the goodness of fit of the regression model.

The p -values of the factor B and C were both < 0.0001 (Table 3), which indicated that light intensity and light time were the factors significantly influencing the growth of *S. platensi*. However, sodium bicarbonate concentration and pH had a weaker influence on the growth of *S. platensi* ($p > 0.05$). The p -values of interaction terms AB, AC, AD, BC, BD, and CD were all higher than 0.05, indicating that these interaction terms had lower influence on the growth of *S. platensi*.

Three-dimensional (3D) surface response plots and two-dimensional contour plots can be used to determine the interactions between the factors and their significance in the response (Mohajeri et al. 2010; Yaakob et al. 2019). The 3D response surface plots and contour plots of the model are shown in Figs. 2 and 3, respectively.

As shown in Fig. 2a, b, and c, the influence of sodium bicarbonate concentration on the growth of *S. platensi* was weak. Varying the sodium bicarbonate concentration in the range of 4–7 g/L had no significant effect on the growth of *S. platensi*. This result is consistent with that of single-factor experiment, which also showed that the promotive effect of sodium bicarbonate was not significant above 4 g/L concentration. The influence of pH was not as marked as that of light intensity and light time (Fig. 2c, e, and f), and the pH had no regular influence on the growth of *S. platensi* in the range of 9.0 to 12.0.

As demonstrated in Fig. 2a and e, light intensity had a significant effect on the growth of *S. platensi*, as the OD₅₆₀ increased from 1.12 ~ 1.45 to 1.40 ~ 1.95 when the light intensity was increased from 1530 to 3030 lx. Similar results were found by Liu et al. (2022) and Yaakob et al. (2019), they also found that light intensity significantly influences the growth of microalga through regulating photosynthesis and metabolic pathways.

Figure 2b, d, and f illustrate that light time is one of the factors significantly influencing the growth of *S. platensi*, as the OD₅₆₀ continuously increased in the range of light time between 8 and approximately 24 h. The OD₅₆₀ kept increasing from 1.40 to 1.72 along with the increase of light time from 8 to 16 h at the light intensity of 3030 lx. However, the OD₅₆₀ just increased from 1.72 to 1.95 when the light time was increased from 16 to 24 h, which

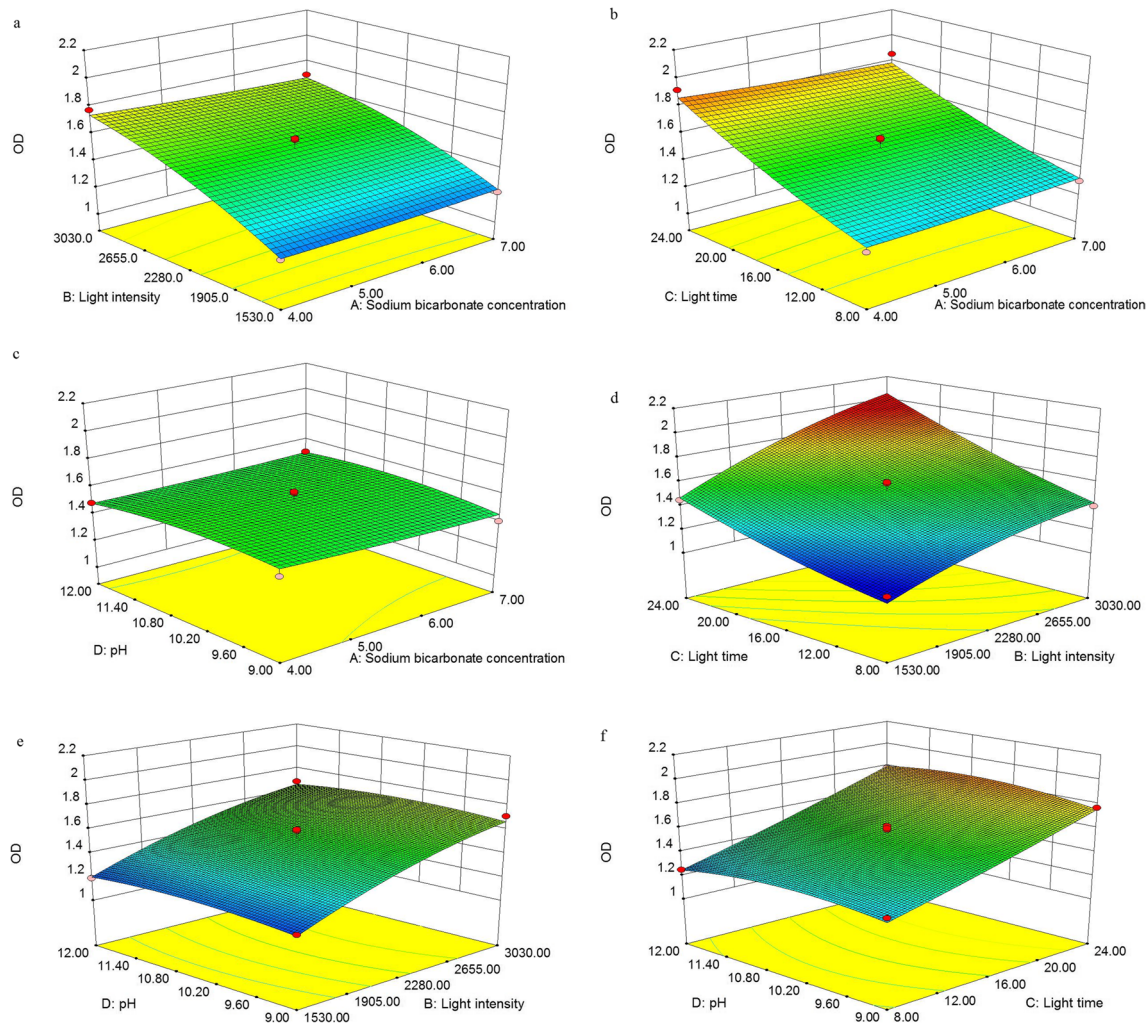


Fig. 2 3D surface response plots of OD₅₆₀ for *S. platensis* cultured in diluted piggery wastewater as a function (**a** A and B, **b** A and C, **c** A and D, **d** B and C, **e** B and D, **f** C and D)

indicated the promotive effect was not significantly as that for 8~16 h. The results confirmed that the light time had direct effect on the growth of *S. platensis*. Liu et al. (2022) also found that the length of light/dark cycle is a significant factor affecting on the growth and nutrient removal activity of *Chlorella*. Eleonora et al. (2012) found that microalgae can efficiently grow under strong light when exposed to an optimal light time, while the dark cycle favors the re-oxidation of electron transporters of the photosynthetic apparatus.

It is well known that elliptical contour plots indicate an optimal interaction between independent variables (Muralidhar et al. 2001). As shown in Fig. 3, there are fewer interactions between the four independent variables because there were no elliptical contour plots. The results of the contour plots confirmed the analysis results of ANOVA, which also showed that the *p*-values were all higher than 0.05 between the interaction independent variables.

The OD₅₆₀ value reached 1.83 (corresponding to 0.56 g dry weight/L) under the optimal growth parameters proposed by Eq. (5), as follows: the dilution ratios of 4, sodium bicarbonate of 7 g/L, light intensity of 3030 lx, light time of 16 h, initial OD₅₆₀ of 0.63, pH of 10.5, and culture time of 8 days. The experimentally verified OD₅₆₀ is very close to the prediction value of 1.68, which confirms the validity of the proposed model. The results indicated that the proposed model could be used to evaluate the growth of *S. platensis* after cultivation in diluted piggery wastewater.

Analysis of the chemical compositions of *S. platensis* biomass In order to evaluate the feasibility of cultivating *S. platensis* in diluted piggery wastewater to produce biomass as animal feedstock, the elemental compositions and chemical compositions (sugar, crude lipid, and total chlorophyll) were analyzed, and the results are shown in Tables 4 and 5, respectively.

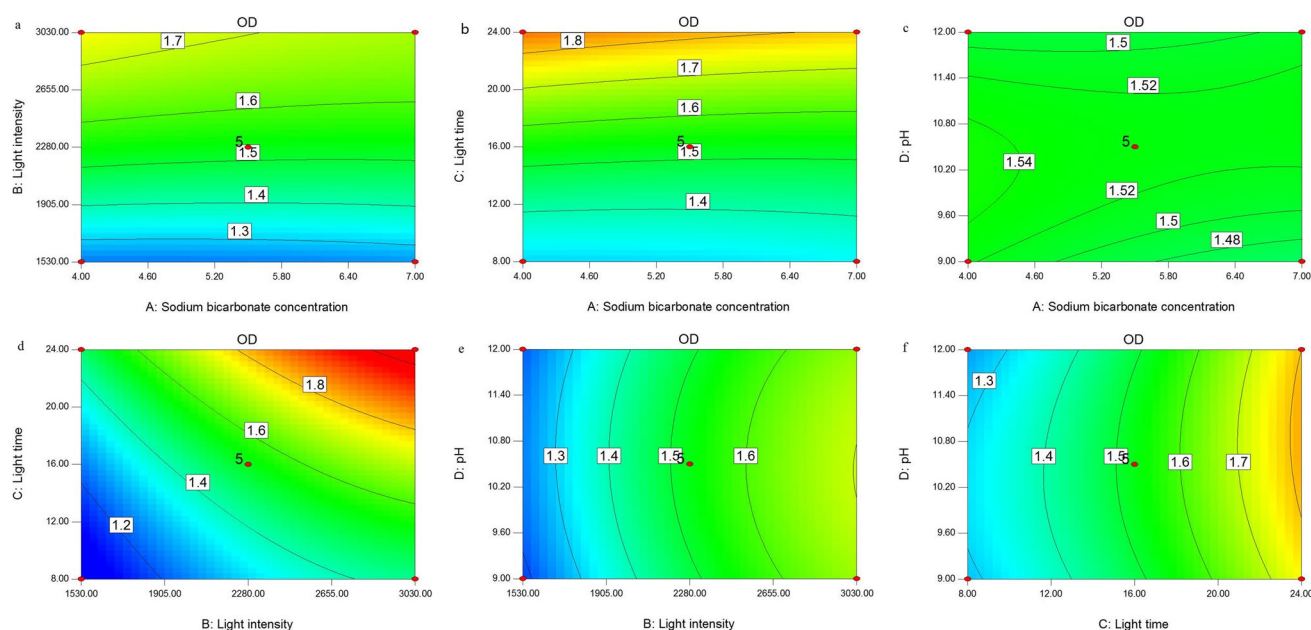


Fig. 3 Contour plots of OD560 for *S. platensi* cultured in diluted piggery wastewater as a function (a A and B, b A and C, c A and D, d B and C, e B and D, f C and D)

As described in Table 4, the contents of C, N, H, and S in the two *S. platensi* biomass groups both followed the order of $C > N > H > S$. The protein contents were 64.96% and 43.89%, respectively, which were calculated according to the N content. The results of elemental analysis were similar to that of Ortega-Calvo et al. (1993), who reported that the content of C, N, H, and S (g/100 g dry biomass) of three *S. platensi* strains were in the range of 51.7 ~ 53.5, 10.66 ~ 12.45, 7.48 ~ 7.92, and 0.63 ~ 0.97, respectively. The differences in elemental composition between the two *S. platensi* biomass groups likely result from the adaptation within its biomass to the variation in nutrient composition in the two media (Whitton et al. 2016; Yaakob et al. 2019).

Table 5 shows that the crude lipid content of *S. platensi* cultured in the wastewater (9.4%) is very close to that cultured in common medium (10.3%). The crude lipid content in the present study was very close to that of Rahim et al. (2021), they reported the lipid content of *S. platensi*

cultured in the well water-wood ash medium was about 9.25%. Tokusoglu and Unal (2003) found that the lipid content of *S. platensi* varies with the microalgae strain, and the lipid content of three strains was between 7.09 and 8.03%. Markou et al. (2012) found that the lipid content of *S. platensi* cultured in the zarrouk medium, synthetic human urine, and real human urine was 7.8%, 17.2%, and 19.8%, respectively. These results showed that the lipid content of *S. platensi* is directly influenced by the nutrient composition of the culture medium. The difference among the current and above works should mainly come from the difference of the microalgae strains and nutrient compositions.

The chlorophyll a of the two biomass groups was 6.626 and 6.41 mg/g, respectively. The chlorophyll a contents of the *S. platensi* cultured in the common medium and wastewater were both slightly lower than the average values of three strains of *S. platensi* at 8.97 mg/g (Rahim et al. 2021).

High performance liquid chromatography was used to analyze the total sugar contents and composition of monosaccharides of the two biomasses. The total sugar contents of *S. platensi* cultured in the common medium and diluted piggery wastewater was 5.38% and 4.18%, respectively. The monosaccharide profiles analyses showed that *S. platensi* total sugar was mainly composed of mannose, ribose, rhamnose, glucuronic acid, galacturonic acid, glucose, galactose, arabinose, and fucose. These data were consistent with previous studies (Chaiklahan et al. 2013; Lee et al. 2012; Mouhim et al. 1993; Ortega-Calvo et al. 1993). Glucose was the most abundant monosaccharide in the two biomasses, which is

Table 4 Elemental compositions of *S. platensi* cultured in common medium and piggery wastewater

Sample	N%	C%	H%	S%
<i>S. platensi</i> cultured in the common medium	10.36	45.4	6.201	0.709
<i>S. platensi</i> cultured in piggery wastewater	7	31.94	5.049	0.475

Table 5 Chemical compositions of the *S. platensi* cultured in common medium and in piggery wastewater

	Unit	Samples		
		<i>Spirulina</i> cultivation in medium	<i>Spirulina</i> cultivation in wastewater	Limits of <i>Spirulina</i> for feed
Protein	%	64.75	43.75	
Heavy metals	Cu(mg/kg)	1.7	27.7	
	Zn(mg/kg)	12.6	246.2	
	As(mg/kg)	0.6	0.9	1
	Cd(mg/kg)	0.03	0.3	0.5
	Hg(mg/kg)	0.01	0.02	0.1
	Pb(mg/kg)	0.05	2.4	6
Crude lipid	%	10.3	9.4	
Total sugar	Mannose (%)	0.038	0.185	
	Ribose (%)	0.326	0.571	
	Rhamnose (%)	0.360	0.952	
	Glucuronic acid (%)	0.109	0.262	
	Galacturonic acid (%)	0.024	0.086	
	Glucose (%)	3.915	1.378	
	Galactose (%)	0.203	0.512	
	Arabinose (%)	0.108	0.101	
	Fucose (%)	0.317	0.193	
	Total sugar	5.381	4.176	
Chlorophyll a	(mg/g)	6.626	6.41	

in good agreement with previous studies (Mouhim et al. 1993; Ortega-Calvo et al. 1993). Rhamnose, ribose, and fucose were the major monosaccharides besides glucose. As depicted as Table 5, there were some differences of monosaccharide composition between the *S. platensi* cultured in the common medium and diluted piggery wastewater. The glucose contents of the *S. platensi* cultured in the common medium and wastewater were 3.92% and 1.38%, respectively. Besides the content of glucose of the former was higher than the latter, most monosaccharides of the former were lower than that of the latter. The differences between the monosaccharide compositions of the two biomasses should mainly resulted from the different nutrients composition in the two-culture media (Lee et al. 2012). The analysis of the sugar contents and monosaccharides compositions showed that the culture with piggery wastewater had a weaker influence on the accumulation of sugar in *S. platensi*.

The heavy metal content in *S. platensi* cultured in piggery wastewater was analyzed to evaluate the feasibility of it as animal feedstock. As shown in Table 5, the content of Cu (27.7 mg/kg) and Zn (246.2 mg/kg) in the *S. platensi* cultured in diluted piggery wastewater was obvious higher than that cultured in common medium. The cupric and zinc concentration in the wastewater far higher than that in common medium resulted in the accumulation of them in the *S. platensi* biomass cultured in the wastewater through

converting to the components of the cells and adsorption on the cell surface. The content of Zn was obviously higher than that of Cu in both biomass groups, and the ratio of Zn/Cu was about 8.9. This result is consistent with that of Tokusoglu and Unal (2003), who found the ratio of Zn (26.8 mg/kg) to Cu (3.1 mg/kg) in *S. platensi* was about 8.6. The reason is mainly that Zn is vital to the growth of microalgae (Krystian et al. 2015), while cupric ions have a serious inhibiting effect (Li et al. 2018). The analysis of the chemical composition of the *S. platensi* cultured with piggery wastewater showed that the biomass was rich in proteins, crude lipids, mineral substances, chlorophyll a, and polysaccharides. Meanwhile, the heavy metals contents in the wastewater were lower than the limits of *S. platensi* used in animal feed, which indicated the *S. platensi* cultured in piggery wastewater is a safe source of biomass for the production of feedstock.

Removal of nutrients and metal ions

High nutrient removal efficiency (mainly of nitrogen and phosphorus) is one of the dual goals of microalgae-based wastewater treatment (Berner et al. 2015). The removal of TN, TP, COD, Zn, and Cu by the cultivation of *S. platensi* in 4-fold diluted non-autoclaved piggery wastewater for 8 days was studied. The initial concentrations of TN, TP, COD, Zn, and Cu in the diluted wastewater were 67.2 mg/L, 24.3 mg/L,

Table 6 Comparison of nutrient removal efficiency from the piggery wastewater by microalgae with literatures

Microalgae strains	Initial nutrients concentration	Cultivation modes	Wastewater	Removal rate of nutrients	References
<i>Desmodesmus</i> sp. CHX1	NH ₄ -N: 104.6 mg/L TP: 3.15 mg/L	Photoautotrophy in the flasks	Piggery wastewater	NH ₄ -N: 88.26% TP: 95.06%	Luo et al. (2019)
<i>Chlorella zofingiensis</i>	COD: 1900 mg/L TN: 80 mg/L TP: 85 mg/L	Tubular bubble column photobioreactors	Piggery wastewater	COD: 79.84% TN: 82.7% TP: 98.17%	(Zhu et al. 2013)
<i>Chlorella</i> sp. MA1 and <i>Coelastrella</i> sp. KE4	NH ₄ -N: ≈ 400 mg/L PO ₄ -P: ≈ 15 mg/L	Photoautotrophy supplemented CO ₂	Swine wastewater	NH ₄ -N: 35 ~ 53% PO ₄ -P: 96.9 ~ 98%	(Qu et al. 2021)
<i>Dictyosphaerium</i> sp.	NH ₄ -N: 102 mg/L TP: 12 mg/L	Photoautotrophy	Swine wastewater	NH ₄ -N < 50% (8 days) TP < 50% (8 days)	(Wang et al. 2019)
<i>C. vulgaris</i>	TN: 220 mg/L TP: 12 mg/L COD: 700 mg/L	Photoautotrophy	Diluted piggery digestate	TN > 90% TP > 90% COD < 30%	(Franchino et al. 2016)
<i>S. platensi</i>	TN: 67.2 mg/L, TP: 24.3 mg/L COD: 763 mg/L	Photoautotrophy in the flasks	Piggery wastewater	TN: 76% TP: 72% COD: 93.1%	The present study

763 mg/L, 0.75 mg/L, and 0.097 mg/L, respectively. The concentrations of TN, TP, COD, Zn, and Cu of the diluted wastewater reduced to 16.2 mg/L, 6.8 mg/L, 53 mg/L, 0.049 mg/L, and 0.017 mg/L, respectively, after treatment with *S. platensi*. The removal efficiency for TN, TP, COD, Zn, and Cu respectively was 76%, 72%, 93.1%, 93.5%, and 82.5%. These results confirmed the effective removal of nutrients from the piggery wastewater by the cultivation of *S. platensi*.

Table 6 shows the comparison of nutrient removal efficiency from the piggery wastewater by the cultivation of microalgae. The removal efficiency for TN and TP in the present study was higher than that reported by Wang et al. (2019), and lower than that found by Luo et al. (2019), Franchino et al. (2016), and Zhu et al. (2013). However, the removal efficiency for COD was higher than that by Zhu et al. (2013) and Franchino et al. (2016). The above differences mainly result from the differences of microalgae strains used. The nutrient removal ability of *Chlorella* from wastewater has been proved successful in laboratory. However, the difficulty in the separation of *Chlorella* biomass from the water after treatment hampers the scale-up of this technology. On the other hand, the present study showed that the treatment of piggery wastewater with *S. platensi* can effectively remove nitrogen, phosphorus, and COD from the piggery wastewater, and the *S. platensi* biomass produced during the process can act as animal feedstock.

Conclusions

The growth of *S. platensi* was significantly inhibited by 0.5 mg/L Cu²⁺, and sodium bicarbonate was the/a limiting for the cultivation of *S. platensi* in the diluted piggery wastewater. The biomass concentration of *S. platensi* reached 0.56 g/L under the optimized conditions proposed by the response surface method. *S. platensi* cultured in the diluted piggery wastewater contained 43.89% protein, 9.4% crude lipid, 6.41 mg/g chlorophyll a, 4.18% total sugar, 27.7 mg/kg Cu, and 246.2 mg/kg Zn, which indicated that it can serve well as animal feedstock. The removal efficiency for TN, TP, COD, Zn, and Cu was 76%, 72%, 93.1%, 93.5%, and 82.5%, respectively. These results demonstrated that the removal of nutrients from piggery wastewater by the cultivation of *S. platensi* as animal feedstock is feasible. Further studies should be conducted on the optimization of scaling up this treatment technology.

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Data availability The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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